SC Honors Geometry - Nappier
8.2–8.3 Triangle Similarity: AA, SSS, and SAS

Speak **L**ike **A** **M**athematician:

What changes from congruence (Ch. 5) to similarity?

Why does AA need only two pairs of angles?

Three Ways to Prove Triangles Similar

AA Similarity: two pairs of congruent ____________________ make the triangles similar.

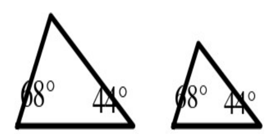
SSS Similarity: all three pairs of corresponding sides are ____________________.

SAS Similarity: two pairs of proportional sides and a congruent ____________________ angle.

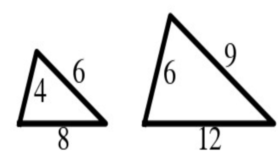
Example 1 – Similar? Say Why (part 1)

For each pair at the right, decide whether the triangles are similar. If they are, name the postulate or theorem.

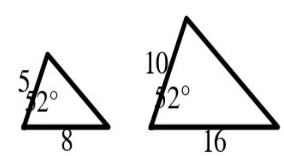
a) First pair (two marked angles each).



b) Second pair (sides 4, 6, 8 and 6, 9, 12).

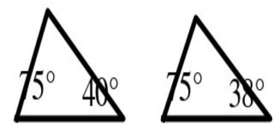


c) Third pair (sides 5, 8 with a 52° included angle; sides 10, 16 with a 52° included angle).

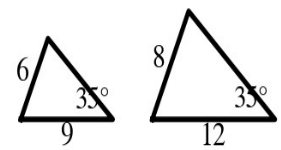


Example 2 – Similar? Say Why (part 2)

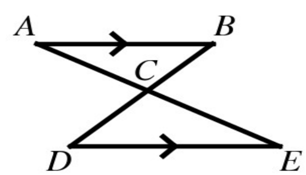
a) Fourth pair (angles 75°, 40° and 75°, 38°).



b) Fifth pair (sides 9, 6 and 12, 8 with a congruent angle NOT between them).



c) Sixth pair ($\overline{AB} \parallel \overline{DE}$ crossing at C).

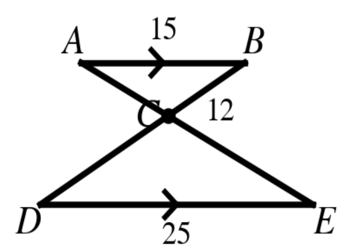


Example 3 – Overlapping Triangles

In the figure at the right, $\overline{AB} \parallel \overline{DE}$, AB = 15, DE = 25, and BC = 12.

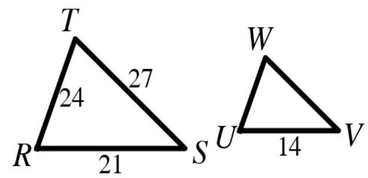
a) Explain why $\triangle ABC \sim \triangle EDC$.

b) Find the scale factor of $\triangle ABC$ to $\triangle EDC$ and the length CD.



Example 4 – Use Similarity to Find Sides

In the figure at the right, △RST ~ △UVW, RS = 21, ST = 27, TR = 24, and UV = 14.



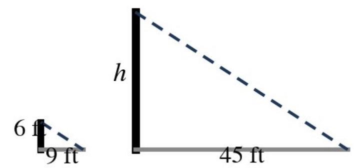
a) Find the scale factor of △RST to △UVW.

b) Find VW and WU.

c) Find both perimeters. What is their ratio?

Example 5 – Indirect Measurement

A 6-ft-tall coach casts a 9-ft shadow. At the same time, the goalpost casts a 45-ft shadow.



a) Why are the two triangles similar?

b) Find the height of the goalpost.

c) Why is this called INDIRECT measurement?